



Biogenic Synthesis and Characterization of Calcium Oxide Nanoparticles Utilizing *Bacillus paramycoides* to Enhance Biofortification in *Abelmoschus esculentus* L.

Farukh Hussan¹ · Somedutta Maity¹ · K. Lakshmi Narayana Reddy¹ · Karuganti Sukumar¹ · Emani Sri Gayathri Reddy¹ · Ramesh Koduru²

Received: 7 January 2026 / Accepted: 4 June 2026

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026

Abstract

Calcium (Ca) is essential for optimal *Abelmoschus esculentus* L. (okra) development, contributing to cell wall fortification, membrane stability, and efficient signal transduction. Adequate Ca supply enhances vegetative vigor, chlorophyll retention, reproductive success, and fruit structural integrity, while deficiency results in impaired growth, reduced fruit set, and tissue breakdown. Improving calcium bioavailability, particularly via nanoscale CaO formulations, can significantly elevate nutrient uptake efficiency and overall crop performance. This study aimed to develop biogenic calcium oxide nanoparticles (CaONPs) synthesized by *Bacillus paramycoides* through an environmentally benign green chemistry route and to investigate their impact on calcium content and growth performance of okra. The synthesized CaONPs were characterized by UV-Vis spectroscopy, Fourier-transform infrared spectroscopy (FTIR), transmission electron microscopy (TEM), and energy-dispersive X-ray (EDX) analysis. The presence of a distinct absorption peak at 380 nm in the UV-Vis spectrum confirmed the formation of CaONPs, while TEM observations indicated that the nanoparticles were spherical in shape with an average diameter of 25 nm. ICP-MS results indicated a substantial enhancement in growth in the treated samples compared to the untreated and market control groups when administered at the same treatment concentration. The yield parameters exhibited a positive correlation with increasing concentrations of CaONPs. The study demonstrates that *Bacillus paramycoides*-derived CaONPs nanoparticles function as a growth stimulant, leading to a significant increase in both growth performance and calcium content in okra. This suggests that there is a relationship between the dosage of CaONPs and their potential for improving agricultural biofortification.

Keywords Green synthesis · Calcium oxide nanoparticles · *Bacillus paramycoides* · *Abelmoschus esculentus* L. · Biofortification · Nano fertilizer

1 Introduction

As the world's population is expected to exceed 9 billion people by 2050, increasing food consumption by around 59–98%, ensuring global food security continues to be a significant concern [1]. Sustainable agriculture practices

that increase crop output while preserving environmental stability and effective resource utilisation are necessary to meet this demand. Despite being crucial for crop production, conventional fertilisation techniques frequently have poor nutrient-use efficiency because of losses from leaching, volatilisation, and soil fixation [2]. The need for better fertiliser delivery systems is highlighted by these inefficiencies, which raise production costs and harm environmental conditions [3].

A crucial macronutrient, calcium (Ca) has basic structural and regulatory functions in the growth and development of plants. It functions as a crucial secondary messenger in plant signalling networks, preserves membrane integrity, controls enzyme activity, and stabilises cell walls through

✉ Farukh Hussan
farukh@biofactor.in

¹ AADHAAR, DSIR certified R and D Lab, Biofac Inputs Private Limited, Hyderabad, Telangana 502 325, India

² Department of Biochemistry, Yogi Vemana University, Kadapa 516005, AP, India

calcium pectate interactions [4]. Calcium is mostly static in the phloem and is mostly carried by the transpiration stream, in contrast to highly mobile nutrients like nitrogen or potassium [5]. Because of this, even in soils with sufficient calcium levels, quickly growing tissues like root tips, blooms, and developing fruits sometimes experience localised calcium deficit. Inadequate calcium delivery can cause physiological diseases such as blossom-end rot and fruit tissue degradation in crops like *Abelmoschus esculentus* L. (okra), interfere with pollen tube formation, increase membrane leakage and oxidative stress, and hinder cell division and elongation [6–8].

Conventional calcium fertilisers, such lime, gypsum, and calcium nitrate, are frequently used to treat calcium deficiencies, but their low foliar penetration, weak solubility, and soil fixation frequently restrict their efficacy [9]. Repeated high-dose treatments are often required due to these limitations, which may change the chemical balance of the soil and lower overall nutrient-use efficiency. As a result, increasing the amount of calcium that plants receive has emerged as a key goal in sustainable nutrition management [7].

The use of nanotechnology has shown promise in increasing fertiliser efficiency. High surface area, improved reactivity, and regulated nutrient release are some of the special physicochemical characteristics of metal and metal oxide nanoparticles. Calcium oxide nanoparticles (CaONPs) in particular have garnered interest as possible nano-fertilizers due to their nanoscale size, which makes it easier for them to penetrate plant surfaces like stomatal apertures and cuticular holes, improving nutrient availability [10]. CaO can quickly hydrate to $\text{Ca}(\text{OH})_2$ in aqueous or biological conditions, producing bioavailable Ca^{2+} ions that can take part in physiological processes related to membrane function and cell wall stabilisation [11, 12]. However, the safe use of chemically produced nanoparticles in agricultural systems may be restricted due to their potential for instability, aggregation, or phytotoxicity [13].

An alternate strategy that overcomes a few of these constraints is the biological production of nanoparticles. Through extracellular biomolecules, including proteins, enzymes, polysaccharides, and organic metabolites that control nucleation, development, and stabilisation, microorganisms can facilitate the production of nanoparticles [14]. The capacity of *Bacillus* species to create metal and metal oxide nanoparticles by ecologically safe methods has been extensively documented [15]. *Bacillus paramycoides* can serve as a biological nanofactory in this situation, promoting the production of CaONPs and offering biomolecular capping agents that enhance particle stability and biocompatibility [16].

The commercially significant vegetable crop *Abelmoschus esculentus*, sometimes known as okra, is extensively grown in tropical and subtropical areas. Its nutritional makeup, which includes dietary fibre, vitamins, minerals, and antioxidant chemicals, makes it valuable. Despite its significance, okra output is frequently limited by environmental challenges and nutritional imbalances, with calcium shortage being a particularly limiting factor impacting growth, reproductive development, and fruit quality [8].

Therefore, the current work uses *Bacillus paramycoides* to examine the microbial manufacture of calcium oxide nanoparticles and assesses their potential use in enhancing *Abelmoschus esculentus* growth, physiological efficacy, and yield development. This paper attempts to investigate a different approach for improving calcium delivery and promoting sustainable agricultural production by combining microbial nanotechnology with nutrient management [17].

2 Materials & Methods

2.1 Chemical & Culture Media

The study utilized chemical reagents obtained from SRL, India, while the culture media were acquired from HiMedia, India.

2.2 Molecular Characterisation

For definitive taxonomic identification, the bacterial isolate was analyzed by PCR-based amplification and sequencing of the 16 S rRNA gene. The resulting partial 16 S rRNA gene sequence exhibited the highest similarity to *Bacillus paramycoides* and has been deposited in the GenBank database under accession number MW664369.1.

2.3 Synthesis of Biogenic CaONPs and Characterization

Nutrient broth (HiMedia, Mumbai) was made ready further, autoclaved, and allowed to cool to approximately 45–50 °C. A loopful of a 24-h culture of *B. paramycoides* was inoculated into the sterile broth, then kept in an orbital shaker for 24 h at 37 °C, maintained at 100 rpm (Oribtek LXD). Following incubation, the bacterial culture was centrifuged for 10 min at $10,000 \times g$, and the supernatant was separated for CaO nanoparticle synthesis. A total of 300 mL of the collected supernatant was transferred into a conical flask, to which 150 mL of a 1 mM aqueous calcium nitrate solution (50% v/v) was added. The reaction mixture was incubated at 37 °C in bright light while being continuously shaken at 100 rpm in an orbital shaker. By taking 2 ml aliquots of the

reaction liquid and determining the absorbance using a UV-visible spectrophotometer in the 300–500 nm wavelength range, the formation of CaO nanoparticles was monitored at regular intervals. The reaction mixture was centrifuged for 30 min at $10,000 \times g$ in a REMI cooling centrifuge for further characterisation. After twice centrifuging the resultant nanoparticles with sterile distilled water, they were allowed to air dry at room temperature for a whole night.

The CaO nanostructures were characterized using multiple analytical techniques. UV-Visible spectroscopic analysis was performed in the wavelength range of 180–900 nm using a Lambda 750 spectrometer (PerkinElmer). Transmission electron microscopy (TEM) analysis was carried out using an FEI Tecnai G2S-Twin microscope operated at an accelerating voltage of 200 kV to examine the morphology and size of the nanoparticles. Fourier transform infrared (FTIR) spectroscopy (Nicolet, Impact-410 model) was employed to determine which chemical functional groups are present on the CaONPs surface. The hydrodynamic diameter and size distribution of the nanoparticles were determined by dynamic light scattering (DLS) using a Zeta-sizer Nano ZS (Malvern) at 24.9 °C. For analysis, 1 mg of the air-dried nanoparticle sample was dispersed in sterile distilled water. Measurements were recorded over a particle size range of 0.11000 nm.

Inductively Coupled Plasma-Mass Spectrometry (ICP-MS) was used to assess the elemental composition of the synthesised CaO nanoparticles (CaONPs) and calcium presence in plant tissues. A combination of concentrated nitric acid (HNO_3) and hydrogen peroxide (H_2O_2) was used to digest around 10 mg of CaONPs for nanoparticle characterisation. The mixture was heated under controlled conditions until total dissolution was attained. Before analysis, the digested solution was filtered and diluted with ultrapure deionised water to a predetermined volume.

Samples (fruit tissues) were oven-dried at 65–70 °C to constant weight, completely cleaned with deionised water to eliminate surface impurities, and then ground into a fine powder for plant tissue analysis. In a closed-vessel microwave digestion system, around 0.5 g of the dried sample was broken down using concentrated HNO_3 and a small quantity of H_2O_2 until full extraction was obtained. Before analysis, the digested solutions were chilled and diluted with ultrapure deionised water to a specified final volume. A calibrated device with approved multi-element standard solutions for external calibration was used to perform ICP-MS experiments. Standard solutions within the expected concentration range were used to create calibration curves. Reagent blanks, duplicate samples, and standard verification solutions were used in quality control processes to guarantee analytical precision and accuracy. The fresh weight basis

(mg kg^{-1} DW) was used to express the calcium concentrations in plant tissues.

2.4 CaONPs Treatment

Okra plants were divided into three different experimental treatment groups: okra untreated as the negative control, distilled water (T1; 10 mL/L), okra treated with a commercially available market product (T2; 10 mL/L), and okra treated with calcium oxide nanoparticles (CaONPs) (T3; 10 mL/L). The elemental calcium content was 25 mg L^{-1} (25 ppm Ca). Applying 35 mg L^{-1} CaO nanoparticles in the final solution was based on the stoichiometric content of CaO (71.4% Ca). The standard suspension was made at 3.5 g L^{-1} . All treatment solutions were prepared using distilled water as the solvent. A hand sprayer was used to apply the prepared solutions as foliar sprays until the leaf surfaces were uniformly moist. To reduce experimental error and ensure accurate comparisons between treatments, the study employed a Randomised Block Design (RBD) with three replicates. Two plants were kept in each pot with the specified growth media, and each treatment comprised two plants. Three months was the whole timeframe of the experiment, which gave enough time to see how the treatments affected the development of the plants. Foliar application of treatment solutions was done once a week. The spraying began after seed germination, when the seedlings were established, and continued regularly throughout the experimental period. A consistent spraying strategy was used to ensure that the foliage was evenly covered. The amount of sprays sprayed corresponds to the weekly application schedule followed during the three-month testing period. The plants were maintained in a greenhouse under controlled environmental conditions to ensure uniform growth and minimize external variability. The temperature within the greenhouse was kept at about 26 °C, with a relative humidity of about 75%. Light settings were adjusted to maintain an intensity of 1200 to 1500 lux.

3 Results & Discussion

Transmission electron microscopy (TEM) was utilized to look into the morphology, particle size distribution, and crystallinity of the synthesised CaONPs. The TEM images reveal the successful formation of uniformly spherical nanoparticles that are well dispersed, with smooth surfaces and a narrow size distribution. Minimal agglomeration is observed, indicating effective regulation of nucleation and growth throughout the synthesis process. The narrow size distribution further confirms the high level of particle uniformity, with an average particle size of approximately ~25 nm,

suggesting homogeneous nanoparticle formation (Fig. a-c). Overall, TEM analysis confirms the successful formation of uniformly spherical, nanoscale CaONPs with high crystallinity and a narrow size distribution. The spherical geometry of the nanoparticles minimizes membrane damage while limited agglomeration enhances their dispersion on leaf surfaces, and together with well-controlled morphology and particle size this leads to increased leaf residence time, higher absorption efficiency, enhanced surface reactivity, and could improved functional performance, making CaONPs particularly advantageous for biofortification applications where high reactivity and bioavailability are essential for effective nutrient delivery.

Bacillus paramycoides plays an essential role in the surface modification of calcium-based nanoparticles synthesized through a biogenic approach. Biomolecules secreted by the bacterium, such as proteins, peptides, polysaccharides, and organic acids, interact with the nanoparticle surface and act as intrinsic capping and stabilizing agents. These interactions lead to the incorporation of surface functional groups, including hydroxyl (-OH), amine (-NH₂), carboxyl (-COOH), and carbonyl (-C=O), as verified by FTIR analysis.

The FTIR spectrum of the CaONPs particles exhibited characteristic absorption bands corresponding to various functional groups. Peaks observed at 3304 cm⁻¹ were assigned to O-H stretching vibrations of alcohol groups, while the band at 2349 cm⁻¹ corresponds to O=C=O stretching. The absorption at 2113 cm⁻¹ is attributed to C≡C

stretching of alkyne groups, and the peak at 1632 cm⁻¹ is associated with C=C stretching of alkenes. Additional bands at 1339 and 1111 cm⁻¹ were assigned to S=O stretching and C-O stretching of secondary alcohols, respectively. Notably, the sharp band at 3304 cm⁻¹ indicates the presence of -OH groups related to Ca(OH)₂ formation (Fig. 1d), which likely arises from the attributed to surface hydroxylation and adsorbed water molecules. For CaO nanoparticles exposed to ambient conditions, these surface hydration effects have been extensively documented. A small band seen between 2113 may tentatively correlate to C-C stretching vibrations, which might result from secondary biomolecules or trace organic compounds that *Bacillus paramycoides* produces during the biogenic synthesis process [18].

The detected surface functional groups such as -OH associated with Ca(OH)₂ formation and hydration layers, -NH₂ from protein or peptide capping, -COOH from organic acid interactions, and metabolic residues including -C=O, -C=C, and -S=O-create a well-defined nano-bio interface that suppresses nanoparticle agglomeration, improves colloidal dispersion, enhances adhesion to the leaf cuticle, enables controlled Ca²⁺ release, and reduces oxidative and ionic toxicity, while biomolecule-mediated regulation of nucleation and growth yields nanoparticles with uniform morphology and well-controlled size, collectively enhancing calcium reactivity and bioavailability and making *Bacillus paramycoides* mediated synthesis an environmentally friendly approach for producing biocompatible, functionally active calcium nanoparticles suited for efficient

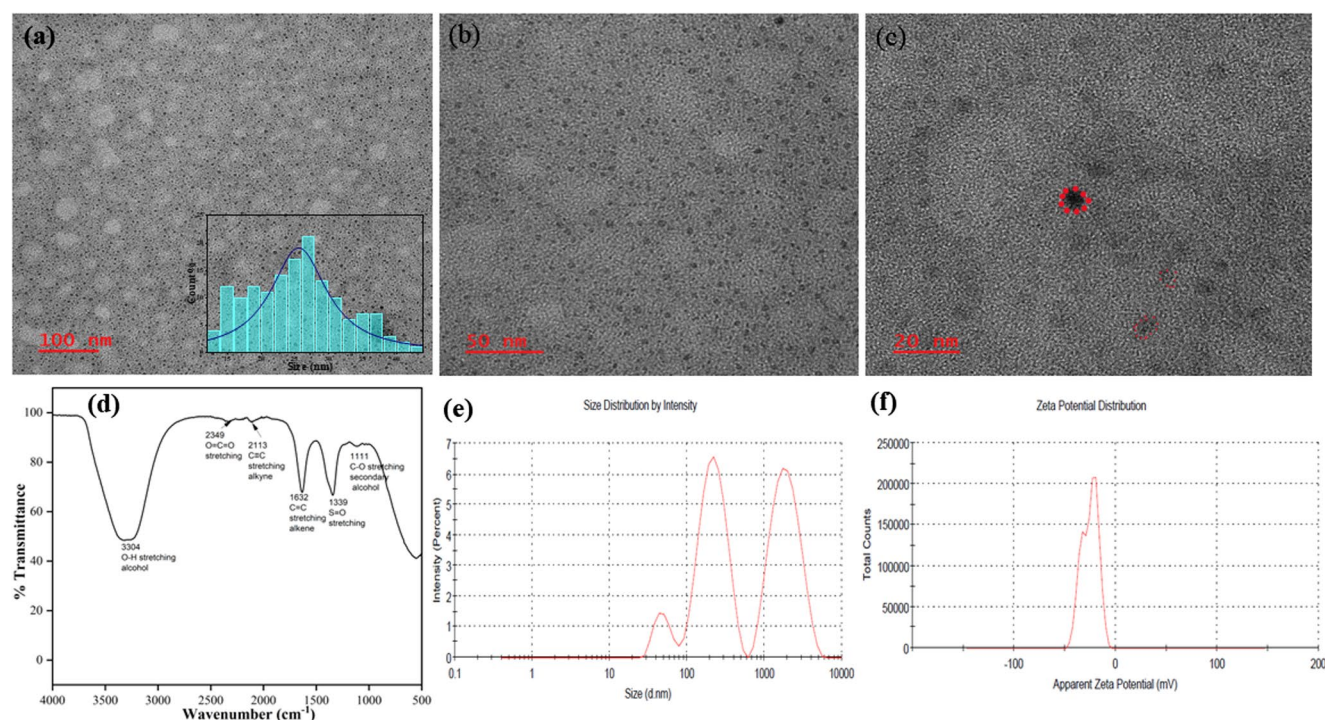


Fig. 1 (a-c) The transmission electron micrograph (TEM) of calcium oxide nanoparticles (d) FTIR data (e) DLS (f) Zeta potential of CaONPs

biofortification and agricultural nutrient delivery. The synthesized CaONPs showed many particle size populations with peak values at 49.86 nm, 229.1 nm, and 1995 nm, as well as a small population at 8.91 nm, according to dynamic light scattering (DLS) analysis (Fig. 1e). A substantially polydisperse colloidal system comprising both primary nanoparticles and aggregated structures is indicated by the measured polydispersity index (PDI) of 0.389. The predominant nanoscale population, which is centered at 49.86 nm, is similar to previously reported CaO nanoparticle sizes and typically agrees with the particle diameters shown in TEM images. However, given the strong surface reactivity and propensity of CaO nanoparticles to agglomerate in aqueous environments, the presence of larger size populations in the DLS profile indicates partial aggregation of nanoparticles in suspension [19].

The zeta potential of the CaONPs, measured in the range of -21.4 to -34.0 mV (Fig. 1f), demonstrates moderate to high electrostatic and colloidal stability of the calcium oxide nanoparticle suspension, indicating strong resistance to aggregation and sustained dispersion under application conditions. It is hypothesized that the pronounced negative surface charge arises from effective biomolecule capping derived from *Bacillus paramycoides*, which could stabilize the nanoparticles and modulate their interactions with biological interfaces. Furthermore, it may be hypothesised that the negatively charged CaONPs could exhibit affinity toward positively charged regions of the plant cell wall and calcium-binding pectin domains, which may contribute to their association with plant tissues. However, further experimental studies are required to confirm these interactions and their physiological significance. When combined with their nanoscale dimensions, this favourable surface chemistry may significantly improve foliar adhesion, residence time. It could enhance calcium uptake efficiency, thereby

supporting more effective calcium delivery and providing a mechanistic explanation for the observed enhancements in plant growth and yield reported in this biofortification study.

X-ray diffraction (XRD) was used to analyze the crystalline structure of the synthesized nanoparticles. The resulting diffraction pattern is shown in Fig. 2a. The standard JCPDS data (File No. 00-004-0777) and the principal diffraction peaks agreed well, suggesting that the primary crystalline phase is CaO with a cubic crystal structure. A few low-intensity peaks attributed to CaCO_3 and potentially Ca(OH)_2 were also seen, along with the typical CaO reflections. Because CaO is hygroscopic and reactive, the occurrence of these minor phases may be linked to partial surface carbonation and hydration of CaO nanoparticles following contact with ambient moisture and carbon dioxide during sample handling or storage. Because of their large surface area and increased surface reactivity, CaO nanoparticles have shown similar behaviour in the published journal. However, because no quantitative phase analysis was done, the interpretation of these later phases is still qualitative [20].

UV-Visible spectroscopy research provided additional evidence for *Bacillus paramycoides* involvement in the synthesis and surface functionalisation of Calcium oxide nanoparticles. The biosynthesized calcium oxide nanoparticles' UV-Vis spectra showed a clear absorption band at around 380 nm (Fig. 2b), showing electronic transitions impacted by surface modification and validating the creation of CaO nanoparticles. This absorption behaviour implies that bacterial biomolecules were involved in the synthesis process. These biomolecules probably work as stabilising and reducing agents, helping to stabilise and functionalise the nanoparticles. Thus, plasmonic phenomena cannot be responsible for the observed absorption band. However, electronic transitions from the valence band are linked to the absorption at 380 nm. Thus, plasmonic phenomena cannot

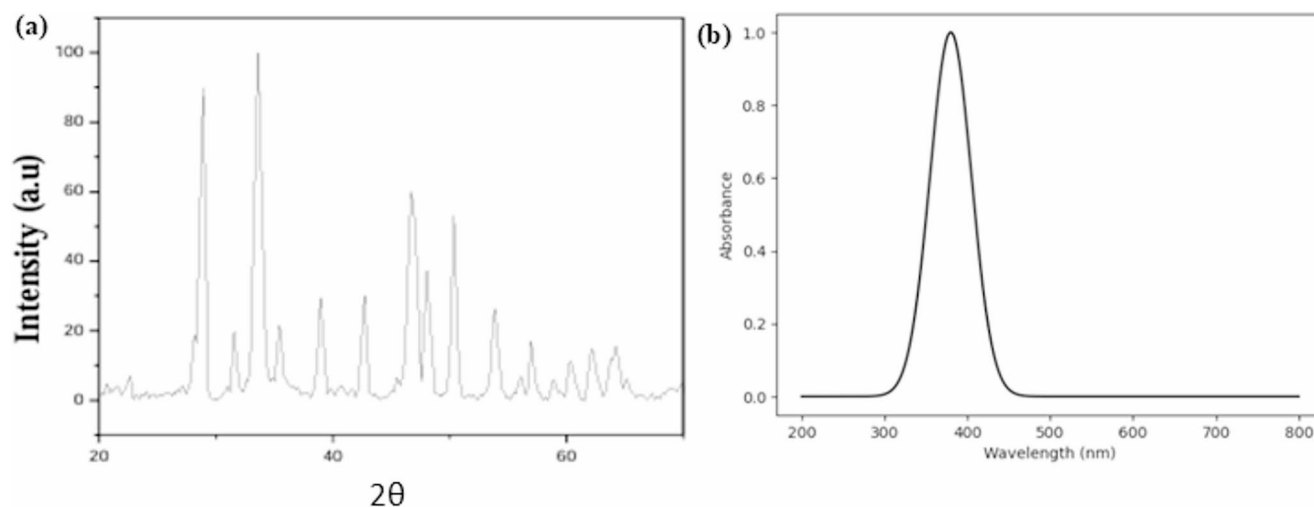


Fig. 2 (a) The X-ray diffraction (XRD) spectrograph of calcium oxide nanoparticles. (b) UV-vis spectrum of calcium oxide nanoparticles

be responsible for the observed absorption band. Rather, defect-related states within the band gap and electronic transitions from the valence band to the conduction band are linked to the absorption at 380 nm. Oxygen vacancies and surface defect states are examples of structural flaws in metal oxide nanoparticles that might introduce localised energy levels assisting in near UV absorption [21]. Additionally, because of enhanced surface effects and potential quantum confinement, the nanoscale size of the particles may affect the optical characteristics [22].

3.1 Foliar Application of Calcium Oxide Nanoparticles

Fertilizer application in crop production is commonly achieved through soil or foliar routes; however, foliar application has gained increasing attention because of its operational simplicity, economic advantages, and superior nutrient-use efficiency, particularly in the context of nano-fertilizers and biofortification strategies, where direct and rapid nutrient absorption through leaf tissues bypasses soil-related constraints such as fixation, leaching, and microbial immobilization, and in the specific case of nano-calcium, the small particle size and high surface reactivity enable efficient penetration through stomatal openings and cuticular microchannels, followed by in situ hydration and gradual Ca^{2+} release into the apoplast and cell wall matrix. Calcium may contribute to pectin cross-linking and could support cell expansion and signalling processes in plants. In addition, the gradual nutrient release behavior is expected to better synchronize nutrient availability with physiological demand. Based on these possible mechanisms, it is anticipated that the formulation may assist nutrient translocation, tissue integrity, growth performance, yield, and nutritional quality in edible plant parts, thereby potentially supporting sustainable agricultural productivity [23].

Inductively Coupled Plasma Mass Spectrometry (ICP-MS) analysis revealed significant differences in the concentration of the analyzed element among the various okra (*Abelmoschus esculentus*) treatment groups (Fig. 3a-c). The recorded calcium concentrations ranged from 10.227 to 20.840 ppm, indicating a pronounced influence of treatment conditions on elemental accumulation.

There were distinct variations in the accumulation of calcium between the treatments (T1, T2, and T3) according to ICP-MS data. With measured values of 10.227, 10.02, and 10.401 ppm DW (average~10.22 ppm), the untreated control (T1) showed the lowest calcium concentration. The calcium content of the T2 treatment, which was given the commercial market product, increased somewhat, measuring 11.928, 11.635, and 12.19 ppm (average~11.92 ppm). On the other hand, the greatest calcium concentrations

were found in okra plants treated with CaO nanoparticles (CaONPs) (T3), with values of 20.84, 20.75, and 20.232 ppm (average~20.61 ppm). According to statistical analysis, there was a significant ($P<0.01$) increase in calcium concentration from T1 to T2, indicating that the commercial treatment somewhat increased the accumulation of calcium in comparison to the control. It is anticipated that the treatments T2 and T3 will exhibit a significant difference ($P<0.0001$), with plants treated with CaONPs exhibiting improved calcium uptake in comparison to the other treatments. The applied therapy greatly improved calcium availability and absorption in plant tissues, as seen by the significantly increased calcium levels in T3 (Fig. 3d). Overall, the findings conclusively demonstrate that T3 accumulated much more calcium than T1 and T2. Because of their nanoscale size and strong surface reactivity, which may promote penetration through the leaf cuticle and stomatal apertures as described for nano-based nutrient formulations, the synthesised CaONPs, when administered by foliar spraying, probably boosted calcium absorption. The effective nano-assisted biofortification is demonstrated by the increased calcium content in plant tissues, showing the potential of CaONPs to improve the mineral nutritional quality of okra. These results imply that developments to address calcium shortage through the biofortification of commonly consumed vegetables might be supported by nano-enabled nutrient delivery systems, which may further contribute to calcium enrichment in crops.

The effects of CaONPs on growth and yield are likely caused by coordinated responses that function at several biological scales. CaONPs may improve calcium pectate cross-linking in the cell wall at the cellular level, stabilising wall structure, boosting membrane integrity, lowering ion leakage, and enhancing buffering against reactive oxygen species. Fruit tissue structures may be strengthened, pollen viability and fertilisation efficiency may be enhanced, and blossom end rot may be less common as a result of these cellular reactions. When taken as a whole, these processes may result in more effective integrated partitioning, stronger sinks, more stable yields, and higher-quality harvested food in terms of nutrition.

4 Conclusion

In this work, *Bacillus paramycoides* was used to successfully synthesise calcium oxide nanoparticles (CaONPs) utilising a biological method. The synthesis and stability of the nanoparticles were verified by characterisation investigations. While dynamic light scattering revealed a hydrodynamic diameter of around 49.86 nm, indicating moderately broad distribution in suspension, TEM

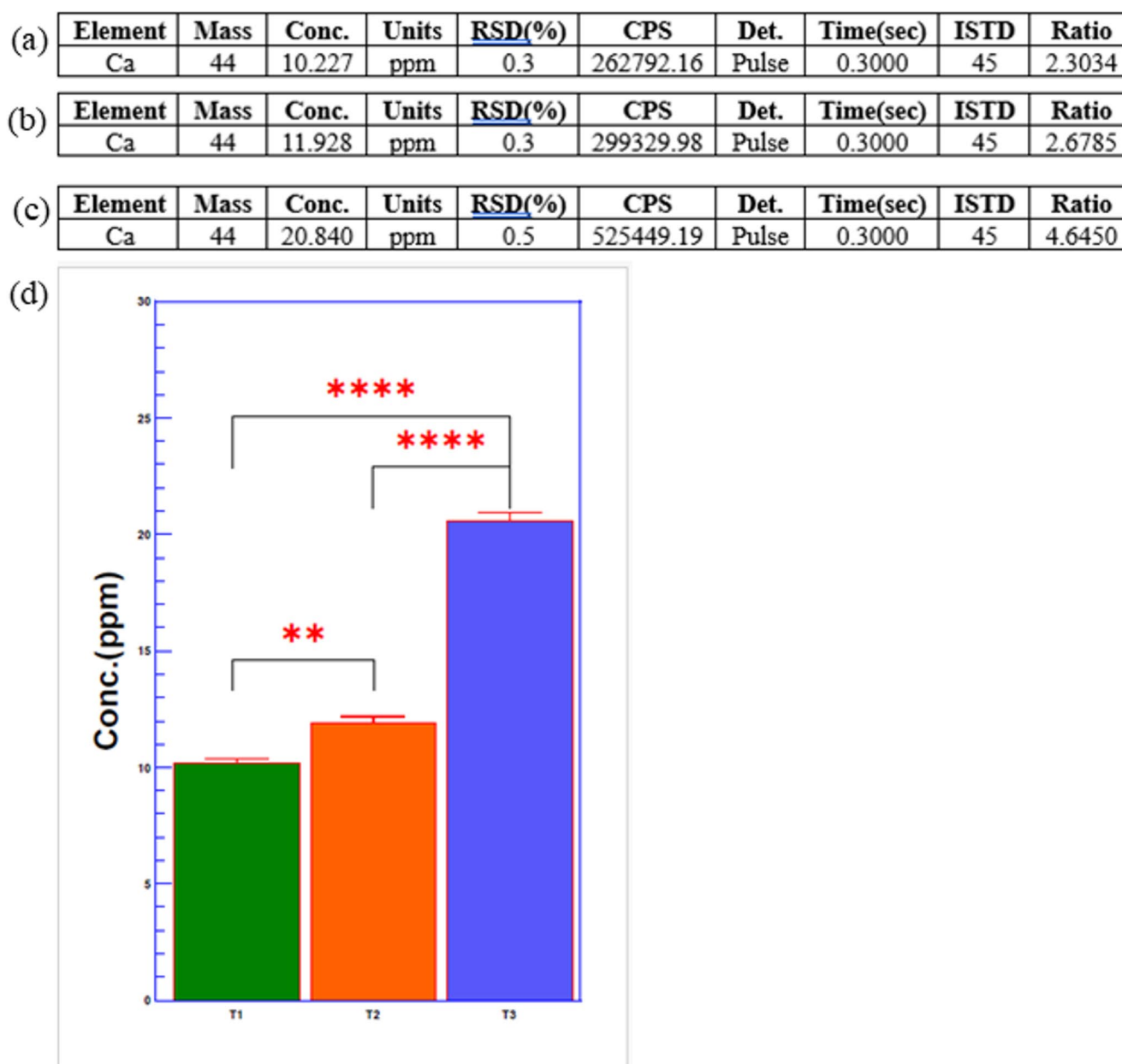


Fig. 3 ICP-MS determination of calcium content in samples following treatment: (a) untreated group (T1), (b) treated with market product (T2) (c) CaONPs (T3) (d) Statistical analysis of three different samples. Each treatment was tested in three replicates, and statistical anal-

ysis was performed using one-way ANOVA followed by Tukey's test. Brackets indicate statistically significant differences, as mentioned below, p -values. The data are presented in the graph as mean $\pm$ standard deviation. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$

investigations revealed nanoscale particles with an average size of about ~ 25 nm. The O-H stretching vibrations are represented by a large absorption band in the ~ 3200 – 3600 cm^{-1} area of the FTIR spectra, while the presence of adsorbed water and calcium hydroxide is indicated by a band about ~ 1640 cm^{-1} due to H-O-H bending and these features, together with the XRD results, suggest that the synthesized material consists of CaO-derived phases with partial hydration and carbonation. The modest colloidal stability shown by zeta potential values (-21.4 and -34.0 mV) was

probably caused by biomolecular capping. UV-visible spectroscopy revealed a distinctive absorption peak at 380 nm, *Abelmoschus esculentus* tissues accumulated more calcium after the synthesized CaONPs were applied. ICP-MS analysis revealed that treated plants had calcium levels of 20.61 ppm when compared to other treatments. These results suggest that CaONPs produced by microbes can be a source of bioavailable calcium and could help plants absorb calcium more effectively. Overall, the work shows that synthesising calcium oxide nanoparticles through microbial processes is

feasible and offers initial proof of their possible use in agricultural nutrition management. To fully comprehend their agronomic efficacy, environmental behaviour, and practical usefulness in sustainable agriculture, more field research and long-term evaluations will be required.

Acknowledgements F. H., S. M., K. L. N. Reddy, ESG Reddy, R.K. and K. S. thank Prof. Dibakar Das, School of Engineering Sciences and Technology, University of Hyderabad, for providing access to the nanomaterial characterization facilities.

Author Contributions Farukh Hussan: Writing review & editing, Ideation, Software, Resources, Methodology, Formal analysis, Data curation. Samedutta Maity: Writing-original draft, review & editing, Characterization. K. Lakshmi Narayana Reddy: Ideation, Visualization, Funding acquisition, Resources, Methodology, Investigation, Formal analysis. Karuganti Sukumar: Writing -review & editing, Visualization, Software, Resources, Investigation, Formal analysis, Data curation. Emani Sri Gayathri Reddy: Writing review & editing, Ideation, Characterization. Ramesh Koduru: Writing review & editing, Ideation, Characterization.

Funding No funding was received from any agencies.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethics Not applicable.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Competing interests The authors declare no competing interests.

References

- Gupta, S., Kant, K., Kaur, N., Jindal, P., Ali, A., & Naeem, M. (2025). Nano-Calcium Applications in Modern Agriculture: A Review. *Plant Nano Biology*, 12, 100147. <https://doi.org/10.1016/j.plana.2025.100147>
- Prasad, R., Bhattacharyya, A., & Nguyen, Q. D. (2017). Nanotechnology in sustainable agriculture: recent developments, challenges, and perspectives. *Frontiers In Microbiology*, 8. <https://doi.org/10.3389/fmicb.2017.01014>
- Yadav, N., Garg, V. K., Chhillar, A. K., & Rana, J. S. (2023). Recent Advances in Nanotechnology for the Improvement of Conventional Agricultural Systems: A Review. *Plant Nano Biology*, 4, 100032. <https://doi.org/10.1016/j.plana.2023.100032>
- Cakmak, I. (2013). Magnesium in Crop Production, Food Quality and Human Health. *Plant And Soil*, 368(1–2), 1–4. <https://doi.org/10.1007/s11104-013-1781-2>
- Wahab, A., Batool, F., Muhammad, M., Zaman, W., Mikhlef, R. M., Qaddoori, S. M., Ullah, S., Abdi, G., & Saqib, S. (2024). Unveiling the Complex Molecular Dynamics of Arbuscular Mycorrhizae: A Comprehensive Exploration and Future Perspectives in Harnessing Phosphate-Solubilizing Microorganisms for Sustainable Progress. *Environmental And Experimental Botany*, 219, 105633. <https://doi.org/10.1016/j.envexpbot.2023.105633>
- Kour, J., Khanna, K., Singh, A. D., Dhiman, S., Bhardwaj, T., Devi, K., Sharma, N., Ohri, P., & Bhardwaj, R. (2023). Calcium's Multifaceted Functions: From Nutrient to Secondary Messenger during Stress. *South African Journal of Botany*, 152, 247–263. <https://doi.org/10.1016/j.sajb.2022.11.048>
- Xing, Y., Xie, Y., & Wang, X. (2025). Enhancing soil health through balanced fertilization: A pathway to sustainable agriculture and food security. *Frontiers In Microbiology*, 16. <https://doi.org/10.3389/fmicb.2025.1536524>
- Elkhalifa, A. E. O., Alshammari, E., Adnan, M., Alcantara, J. C., Awadelkareem, A. M., Eltoum, N. E., Mehmood, K., Panda, B. P., & Ashraf, S. A. (2021). Okra (*Abelmoschus Esculentus*) as a Potential Dietary Medicine with Nutraceutical Importance for Sustainable Health Applications. *Molecules*, 26(3), 696. <https://doi.org/10.3390/molecules26030696>
- Mohammed, H. Y., Teferra, T. F., & Sime, G. (2024). Indigenous Knowledge and Nutritional and Morphological Characterization of Okra (*Abelmoschus Esculentus* (L.) Moench) Varieties in Western Ethiopia. *Food Sci Nutr*, 12(4), 2537–2550. <https://doi.org/10.1002/fsn3.3936>
- WHITE, P. J. (2003). Calcium in Plants. *Ann Bot*, 92(4), 487–511. <https://doi.org/10.1093/aob/mcg164>
- Nair, R., Varghese, S. H., Nair, B. G., Maekawa, T., Yoshida, Y., & Kumar, D. S. (2010). Nanoparticulate Material Delivery to Plants. *Plant Science*, 179(3), 154–163. <https://doi.org/10.1016/j.plantsci.2010.04.012>
- Liu, R., & Lal, R. (2015). Potentials of Engineered Nanoparticles as Fertilizers for Increasing Agronomic Productions. *Science of The Total Environment*, 514, 131–139. <https://doi.org/10.1016/j.scitotenv.2015.01.104>
- Petra Marschner. (2012). *Marschner's mineral nutrition of higher plants*. Elsevier. <https://doi.org/10.1016/C2009-0-63043-9>
- Hepler, P. K., & Calcium (2005). A Central Regulator of Plant Growth and Development. *The Plant Cell*, 17(8), 2142–2155. <https://doi.org/10.1105/tpc.105.032508>
- Kabir, M. Y., & Díaz-Pérez, J. C. (2025). Calcium Route in the Plant and Blossom-End Rot Incidence. *Horticulturae*, 11(7), 807. <https://doi.org/10.3390/horticulturae11070807>
- Sonowal, S., Koch, N., Sarma, H., Prasad, K., & Prasad R.A. (2023). Review on magnetic nanobiochar with their use in environmental remediation and high-value applications. *J Nanomater*, 2023, 1–14. <https://doi.org/10.1155/2023/4881952>
- Reed, L., & Glick, B. R. (2023). The Recent Use of Plant-Growth-Promoting Bacteria to Promote the Growth of Agricultural Food Crops. *Agriculture*, 13(5), 1089. <https://doi.org/10.3390/agriculture13051089>
- Baran, A., Ghorbanzadeh, V., Doğan, Y., Ahmadian, E., Zulfugarova, P., & Mohamed, A. J. (2026). Green Synthesis of Silver Nanoparticles from *Elaeagnus Angustifolia* Extract: Characterization and Evaluation of Antibacterial and Cytotoxic Properties. *Discover Nano*, 21(1), 2. <https://doi.org/10.1186/s11671-025-04423-3>
- Eram, R., Kumari, P., Panda, P., Singh, S., Sarkar, B., Mallick, M., & Verma, S. (2021). Cellular Investigations on Mechanistic Biocompatibility of Green Synthesized Calcium Oxide Nanoparticles with *Danio Rerio*. *Journal of Nanotheranostics*, 2(1), 51–62. <https://doi.org/10.3390/jnt2010004>
- Khan, U. U., Rehman, K., Gul, K., Munir, S., & Sha, Z. A. (2026). CaO Nanoparticles with Calcite Signatures: Antibacterial Function and Biosafety. *Progress in Chemical and Biochemical Research*, 9(2), 130–143. <https://doi.org/10.48309/pcbr.2026.543134.1462>
- Estrella-Pajulas, L. L., & Gamala, B. J. (2024). I. A Review on the Synthesis, Characterization, and Recent Advancements of

- Visible Light-Activated C-TiO₂ Nanomaterials for Environmental Remediation. *Next Nanotechnology*, 6, 100082. <https://doi.org/10.1016/j.nxnano.2024.100082>
22. Para, M., Tariq, A., Zafar, I., Azmat, M., Ul Ain, Q., Yousaf, W., Badhe, P., Abdullah Alsahli, A., & Dabiellil, F. (2025). Role of green-synthesized and characterization of calcium oxide nanoparticles for sustainable wheat growth in saline environments. *Cogent Food Agric*, 11(1). <https://doi.org/10.1080/23311932.2025.2539906>
23. Adnan, M., Tampubolon, K., Ur Rehman, F., Saeed, M. S., Hayyat, M. S., Imran, M., Tahir, R., & Mehta, J. (2021). Influence of Foliar Application of Magnesium on Horticultural Crops: A Review. *Agrinula: Jurnal Agroteknologi dan Perkebunan*, 4(1), 13–21. <https://doi.org/10.36490/agri.v4i1.109>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.